Donyung Kim

dnkim9970@gmail.com | Suwon, South Korea

RESEARCH OBJECTIVE

My research focuses on multimodal computer vision and 3D scene understanding, with the goal of building systems that reconstruct spaces, localize and track people and objects, and understand human–environment interactions from camera and heterogeneous sensor data.

EDUCATION

Yeungnam University **Gyeongsan, South Korea**

M.S. in Electronic Engineering | Mar 2023 – Feb 2025

- Advanced Visual Intelligence Laboratory (AVIL); supervised by Prof. Sungho Kim.
- Thesis: RecNet: Reinforcement Common Feature Mapping Network for Fast Template Matching in Visible-LWIR Images.
- Research interests: multimodal image matching, infrared vision, 3D reconstruction, visual localization and tracking.
- Cumulative GPA: 4.21 / 4.5

Yeungnam University **Gyeongsan, South Korea**

B.S. in Robotics and Mechanical Engineering | Mar 2017 – Feb 2023

- Cumulative GPA: 3.57 / 4.5

WORK EXPERIENCE

NaviFra **Seoul / Suwon, South Korea**

Robotics Researcher (Full-time) | Sep 2025 – Present

- Research and development in 3D rendering and 6-DoF object tracking for robotics applications.

RESEARCH EXPERIENCE

Multimodal Contactless Sensing Research Center (RLRC) **Feb 2024 – Jun 2025**

Researcher — Patient trajectory tracking and monitoring

- Built a real-time 3D monitoring system by integrating image-based 3D mapping, pose estimation, and multi-camera person re-identification.
- Investigated 3D segmentation by integrating reconstructed 3D maps with 2D segmentation masks.

Hanwha Systems — Intelligent Aiming Sight Project **Mar 2023 – Feb 2025**

Researcher — Real-time EO-LWIR image registration

- Designed a pooling-free CNN and an automatic training strategy for cross-spectral template matching; improved matching success rate by 40%.

Daegu–Gyeongbuk Regional Innovation Platform **May 2024 – Dec 2024**

Researcher — Multimodal human-care mobility sensing and AI

- Developed a real-time monitoring system for robot and worker movement in a smart-factory environment.

Agency for Defense Development Project **Jun 2021 – Dec 2021**

Researcher — Optimal route analysis and real-time small-target tracking

- Developed an integrated IR–radar tracking algorithm for a simulated naval engagement environment; achieved an 87% tracking success rate.

JOURNAL PUBLICATIONS

[1] Donyung Kim, Seungeon Lee, Inho Park, Geonjong Kim, and Sungho Kim, “RecNet: Reinforcement Common Feature Mapping Network for Fast Template Matching in Visible-LWIR Images,” IEEE Access, vol. 12, pp. 195890–195905, 2024. doi: 10.1109/ACCESS.2024.3520169.

[2] Seungeon Lee, Donyung Kim, Inho Park, Geonjong Kim, and Sungho Kim, “Perceptible Lightweight Zero-Mean Normalized Cross-Correlation for Infrared Template Matching,” IEEE Access, vol. 12, pp. 164777–164791, 2024. doi: 10.1109/ACCESS.2024.3492206.

CONFERENCE PUBLICATIONS

International

[3] Donyung Kim and Sungho Kim, “Generation of 3D LWIR Thermal Maps Based on Deep Learning SLAM: Feasibility and Evaluation,” Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications VI, Proc. SPIE 13051, 130510T, 2024. doi: 10.1117/12.3013365.

[4] Donyung Kim and Sungho Kim, “Compact Monocular Camera-based Indoor Object Trajectory Extraction System for Enhanced Monitoring,” 24th International Conference on Control, Automation and Systems (ICCAS), pp. 309–312, Jeju, South Korea, Oct. 2024.

[5] Donyung Kim and Sungho Kim, “Deep Learning Based EO-LWIR Image Registration Obstacles: Survey,” 23rd International Conference on Control, Automation and Systems (ICCAS), Yeosu, South Korea, Oct. 2023.

Domestic

[6] Donyung Kim, Jaeho Kim, and Sungho Kim, “Non-upsampling Mapping Method for Multimodal Template Matching,” Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE), 2024. [in Korean]

[7] Donyung Kim, Otae Jang, Sangho Cho, and Sungho Kim, “RF·IR Signal Simulation and Tracking of Warships and Chaff by Distance and Time,” 35th Workshop on Image Processing and Image Understanding (IPIU), Feb. 2023. [in Korean]

PATENTS

[1] Sungho Kim, Chaitali Bhattacharyya, and Donyung Kim, “Black Ice Detection Method and Computing Device for Performing the Same,” Korean Patent No. 10-2831565, registered Jul. 3, 2025.

[2] Sungho Kim, Donyung Kim, and Younggyun Yoon, “Monocular Camera-based Object Position Tracking Method in an Indoor Environment, Computing Device, and Recording Medium,” Korean Patent Application No. 10-2024-0152824, filed Oct. 31, 2024.

[3] Donyung Kim, Sungho Kim, Inho Park, et al., “Deep Learning-based Visible-LWIR Image Registration Method for an Intelligent Aiming Sight,” Korean patent application, filed Jul. 31, 2024.

HONORS & AWARDS

Gold, 2026 Edison Awards — Engineering & Robotics **2026**
NaviDock Vision Docking System | Optical Sensing & Spatial Intelligence

CES 2026 Innovation Awards Honoree — Robotics **2026**
NaviDock: The World's First Marker-less Vision Docking System

Grand Prize (President's Award), Wish Drone Festival — Autonomous Drone Competition **Sep 2017**
Yeungnam University LINC+ Program

TECHNICAL SKILLS

- Programming: Python, MATLAB, C, Java
- Deep learning: PyTorch, ONNX, TensorFlow Lite (Snapdragon XR2)
- Computer vision: object detection and tracking, multi-view re-identification, multimodal image fusion and matching, 3D reconstruction

LANGUAGE

TOEIC Speaking: 150 (Intermediate High), Mar. 2025